Fiscal Year:	FY 2008	**Task Last Updated:**	FY 08/26/2008
PI Name:	Mourad, Pierre D Ph.D.		
Project Title:	Prototype Testing for Non-Invasive Determination of Intracranial Pressure		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Smart Medical Systems and Technology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	98105	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 NSBRI NNH04ZUU003N Human Health in Space
Start Date:	06/01/2005	**End Date:**	05/31/2010
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/31/2010 (from 5/31/2009) per NSBRI (1/2009)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kliot, Michel (University of Washington)		
Grant/Contract No.:	NCC 9-58-SMS00701		
Performance Goal No.:			
Performance Goal Text:			
	Intracranial pressure (ICP) is a vital determinant of brain function whose monitoring and treatment are central to the successful management of patients with a variety of brain diseases and disorders. There are also indications that changes in cerebral autoregulation - one of the primary physiologial mechanisms for maintaining normal ICP, and detectable via alteration of blood flow into the brain - can be altered during space flight in ways associated with space motion sickness and orthostatic intolerance. Currently, ICP is measured by drilling a hole into the skull and placing a device into one of several places within the cranium. This is clearly not an option available to astronauts in space, who often develop brain edema and its associated increase in ICP. Therefore, a simple, non-invasive method of monitoring ICP (nICP) would allow measurement of this critical parameter in astronauts. We have developed several algorithms for nICP based on several different and novel analysis methods, most of which using data derived from standard diagnostic ultrasound of		

Task Description:	the brain and invasively and continually monitored arterial blood pressure (ABP), and shown it feasible. We have also assembled and tested a device that allows collection of the data necessary to implement in real time, and to further refine this algorithm. Specific Aim #1: to test the feasibility of our approach to determining nICP on humans at several hospitals using invasive ABP (iABP), by collecting additional patient data to be used for refining or the existing algorithm. Specific Aim #2: to perform the same feasibility tests and algorithm refinement with non-invasive ABP (nABP). Therefore, the work proposed here offers to perform steps necessary to bringing to space a non-invasive means of determining intracranial pressure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	1) Non-invasively predicting intracranial pressure (nICP): the patent literature and, to a lesser extent, the scientific literature, are littered with failed attempts to determine intracranial pressure in a non-invasive fashion. These considerable efforts have been motivated by the vital role played by this medical variable in the management of individuals suffering from head trauma, subarachnoid hemorrhage and other sources of intracranial bleeding, as well as brain tumors, hydrocephalus, edema, among other disorders. We have demonstrated in our preliminary work for this proposal as well as in our on-going efforts for this proposal several novel means of predicting nICP that has the best chance to date to capture this critical medical parameter whose current means of measurement requires that a neurosurgeon drill a hole through the skull. 2) In work with our industrial collaborators we have bounded the engineering and scientific requirements that need fulfillment in order to automate the transcranial Doppler process for the purposes of deriving medically useful information from measurements of blood flow rate the middle cerebral artery. Given such a system, this would allow researchers and clinicians to easily and autonomously measure the rate of blood flow into the brain - of intrinsic value and a critical parameter in our approach to the non-invasive determination of intracranial pressure. This system would also allow quantification of the existence and extent of emboli transport into the brain, and the quality of cerebral autoregulation. The monitoring of these measurements plays an important role in guiding the medical and/or surgical care of patients with a variety of neurological disorders (as listed above).
Task Progress:	After its establishment, we have been using our unique the computational and logistical infrastructure necessary to most effectively pursue our human-based studies on the non-invasive determination of intracranial pressure. Regarding logistical infrastructure, we have established and used at four different pre-clinical trial sites (those associated with the University of Washington, University of Pennsylvania, Barrow Neurological Institute, and Columbia University) web-based data entry system that allows all of our clinical sites to transmit the human data they collect in a HIPPA compliant way to the University of Washington for analysis. We collected another 57 data sets from patients for our studies and have processed that data. Our on-going modeling efforts for determining ICP based upon novel statistical measures remain consistent with our initial estimates of the complexity of the particular mathematical models we are using to relate in an empirical fashion readily observable human physiological data to intracranial pressure. Our laboratory-based measurements continue to be consistent with our complementary method of using ultrasound to predict intracranial pressure. With these findings we are ready to finalize, write up and publish our three complementary intracranial pressure algorithm based upon human data and our two complementary intracranial pressure algorithms based upon laboratory data.
Bibliography Type:	Description: (Last Updated: 11/21/2013)
Abstracts for Journals and Proceedings	Mohr B, Macaulay M, Ollos RJ, Marzaban C, Morison D, Illian R, Frederickson RC A, Kliot M, Mourad PD. "Non-invasive determination of intracranial pressure using a variety of ultrasound-based methods." NASA Human Research Program Investigators' Workshop, Houston, TX, February 4-6, 2008. Proceedings of NASA Human Research Program Investigators' Workshop, Houston, TX, February 4-6, 2008. , Feb-2008
Awards	Vaswani PA. "NASA Summer Undergraduate Research Program, June 2008." Jun-2008
Awards	Vaswani PA. "The Barry M. Goldwater Scholarship, August 2007." Aug-2007
Awards	Vaswani PA. "UW President's award for best Sophomore student at the University of Washington, June 2008." Jun-2008
Awards	Kang YA. "Applied Physics Laboratory Pat Hardisty Memorial Annual Scholar Award, November 2007." Nov-2007
Awards	Vaswani PA. "(Engineering) Dean's Research Award; Research Fellowship for Advanced Undergraduates, March 2008." Mar-2008
Awards	Vaswani PA. "Astronaut Scholarship, from the Washington NASA Space Grant, May 2008." May-2008
Awards	Vaswani PA. "NASA Summer Undergraduate Research Program, June 2007." Jun-2007
Papers from Meeting Proceedings	Blizzard DJ, Ollos RJ, Vaswani PA, Yu K, Kolve S, Kliot M, Mourad PD. "Tapping the brain with ultrasound for the non-invasive determination of intracranial pressure." University of Washington's Annual Undergraduate Research Symposium, Seattle, WA, May 2008. University of Washington's Annual Undergraduate Research Symposium, May 2008. , May-2008
Papers from Meeting Proceedings	Vaswani PA, Wright DJ, Ollos RJ, Kliot M, Mourad PD. "Non-invasive determination of intracranial pressure by means of vibroacoustography." University of Washington's Annual Undergraduate Research Symposium, Seattle, WA, May 2008. University of Washington's Annual Undergraduate Research Symposium, May 2008. , May-2008